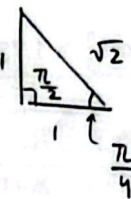
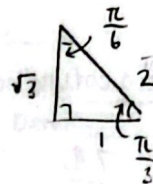


MHF 4U1- Unit 4 Quiz AM - Trigonometry

Name _____

Date _____



Total
15
/17

40 S
9

1. Complete the following conversions in the chart below. (2)

Degree Measure	Exact Radian Measure in terms of π
245°	$245 \times \frac{\pi}{180} = \frac{49\pi}{36}$
$\frac{9\pi}{5} \times \frac{180}{\pi} = 162^\circ$	$\frac{9\pi}{5}$

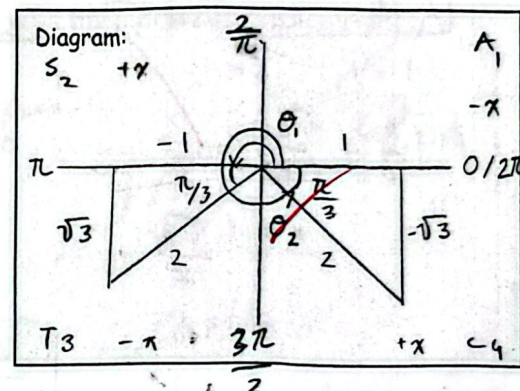
2. Determine the angle(s) of $\sin \theta = \frac{-\sqrt{3}}{2}$ for the domain $0 \leq \theta \leq 2\pi$. Must include a diagram for full marks. (3)

RAA: $\frac{\pi}{3} \rightarrow \frac{\sqrt{3}}{2}$

$\theta_1 = \frac{4\pi}{3}$

$\theta_2 = \frac{5\pi}{3}$

L Q₃, Q₄



3. Use exact values with labelled diagrams to determine $\frac{\cot \frac{\pi}{3} + \csc \frac{3\pi}{4}}{\cos \frac{5\pi}{6}}$. Rationalize. Show all work! (4)

$\frac{\cot \frac{\pi}{3} + \csc \frac{3\pi}{4}}{\cos \frac{5\pi}{6}}$

$\cot \frac{\pi}{3} = \frac{1}{\tan \frac{\pi}{3}} = \frac{1}{\sqrt{3}} \rightarrow \frac{\sqrt{3}}{3}$

$\csc \frac{3\pi}{4} = \frac{1}{\sin \frac{3\pi}{4}} = \sqrt{2}$

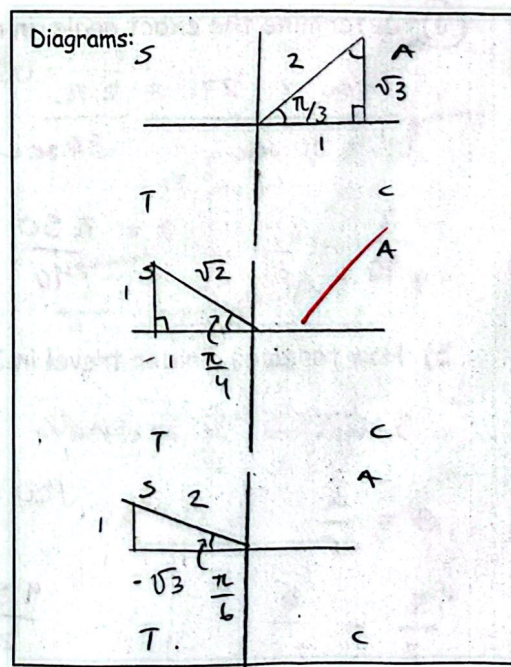
$\cos \frac{5\pi}{6} = -\frac{\sqrt{3}}{2}$

$\frac{\frac{\sqrt{3}}{3} + \sqrt{2}}{-\frac{\sqrt{3}}{2}}$

$\frac{\sqrt{3} + 3\sqrt{2}}{3} \times \frac{2}{-\sqrt{3}}$

$\frac{2\sqrt{3} + 6\sqrt{2}}{-3\sqrt{3}}$

$\frac{2(3) + 6\sqrt{2}}{-3}$
 $\frac{6 + 6\sqrt{2}}{-3}$
 $\frac{-3 - 3\sqrt{2}}{1}$



8.5

$$\rightarrow Q_3 \rightarrow (+)$$

4. Given that $\tan \frac{\pi}{3} \approx 1.732$, use a cofunction identity to evaluate $\cot \frac{7\pi}{6}$ (2)

$$\cot \frac{7\pi}{6} = \tan \frac{\pi}{3}$$

$$\frac{3\pi}{2} - x = \frac{7\pi}{6}$$

$$x = \frac{3\pi}{2} - \frac{7\pi}{6}$$

$$x = \frac{18\pi - 14\pi}{6}$$

$$x = \frac{4\pi}{6} \rightarrow \frac{\pi}{3}$$

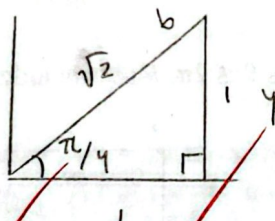
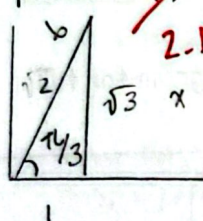
$$\cot \left(\frac{3\pi}{2} - \frac{\pi}{3} \right) = \tan \frac{\pi}{3}$$

$$\cot \left(\frac{9\pi - 2\pi}{6} \right) = \tan \frac{\pi}{3}$$

$$\cot \left(\frac{7\pi}{6} \right) = \tan \frac{\pi}{3}$$

$$\cot \left(\frac{7\pi}{6} \right) = \tan \frac{\pi}{3} \approx 1.732$$

5. Matt is hanging Christmas lights and has his 6 m ladder leaned up against the side of the house. He initially has it raised at an angle of $\frac{\pi}{3}$ to reach the top of the roof, but then lowers it at an angle of $\frac{\pi}{4}$ to hang lights on a window. What is the vertical distance that the ladder moves between the two positions? (3)



$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} = \frac{x}{6}$$

$$x = \frac{6\sqrt{3}}{2}$$

$$x = 3\sqrt{3}$$

$$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} = \frac{y}{6}$$

$$y = \frac{6}{\sqrt{2}}$$

$$\text{distance} = \frac{3\sqrt{3}}{1} - \frac{6}{\sqrt{2}}$$

$$= \frac{3\sqrt{6} - 6}{\sqrt{2}}$$

$$= \frac{3\sqrt{12} - 6\sqrt{2}}{2}$$

$$= \frac{6\sqrt{3} - 6\sqrt{2}}{2}$$

$$= 3\sqrt{3} - 3\sqrt{2}$$

$$= 2\sqrt{3} - \sqrt{2}$$

vertical distance moved

6. A giant Merry-Go-Round has a diameter of 44m and takes 1 minute and 20 seconds to complete a full revolution.

$$r = 22 \text{ m}$$

$$\theta = \frac{a}{r}$$



a) Determine the exact angle, in radians, that a rider will travel 50 seconds. (1)

$$\frac{1 \text{ rev} \times 2\pi}{80 \text{ sec}} = \frac{2\pi}{80 \text{ sec}} \rightarrow \frac{\pi}{40 \text{ sec}}$$

$$\frac{\pi}{40} = \frac{x}{50} \quad x = \frac{\pi 50}{40} \quad x = \frac{5\pi}{4}$$

the exact angle the rider will travel is

$$\frac{5\pi}{4} \text{ radians}$$

b) How far does a rider travel in 3 minutes? (2)

$$3 \text{ min} = 180 \text{ seconds}$$

$$\theta = \frac{a}{r}$$

$$\frac{5\pi}{4} = \frac{a}{22}$$

$$a = \frac{188\pi}{2}$$

$$\frac{180 \times \pi}{40 \text{ sec}} = \frac{9\pi}{2}$$

$$\frac{9.5\pi}{2\pi} \times 44\pi = \frac{1980\pi}{2}$$

the rider will travel 990m

$$180 \times \frac{\pi}{40} = \frac{9\pi}{2}$$

$$\frac{1980\pi}{2} = 990\pi$$